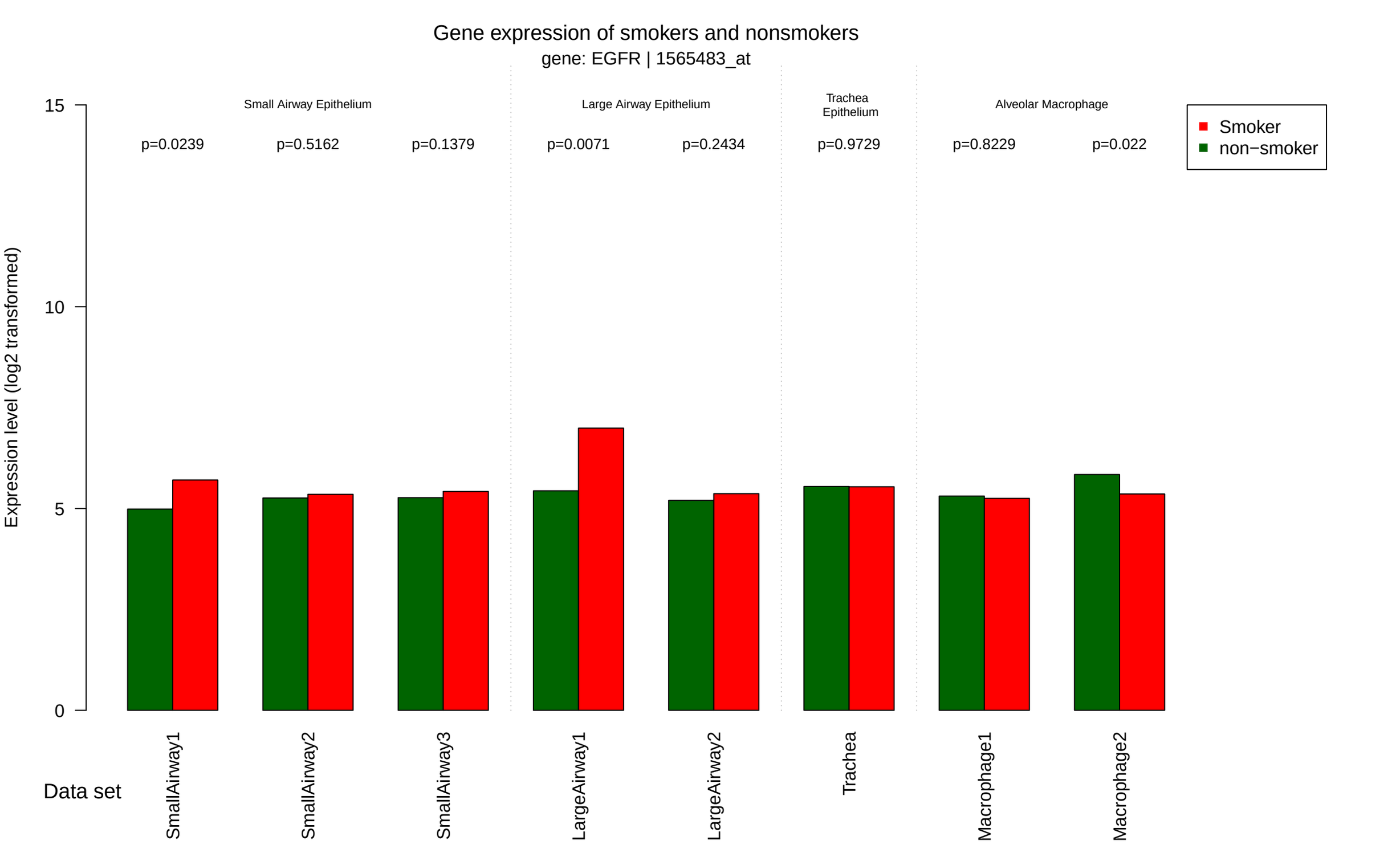
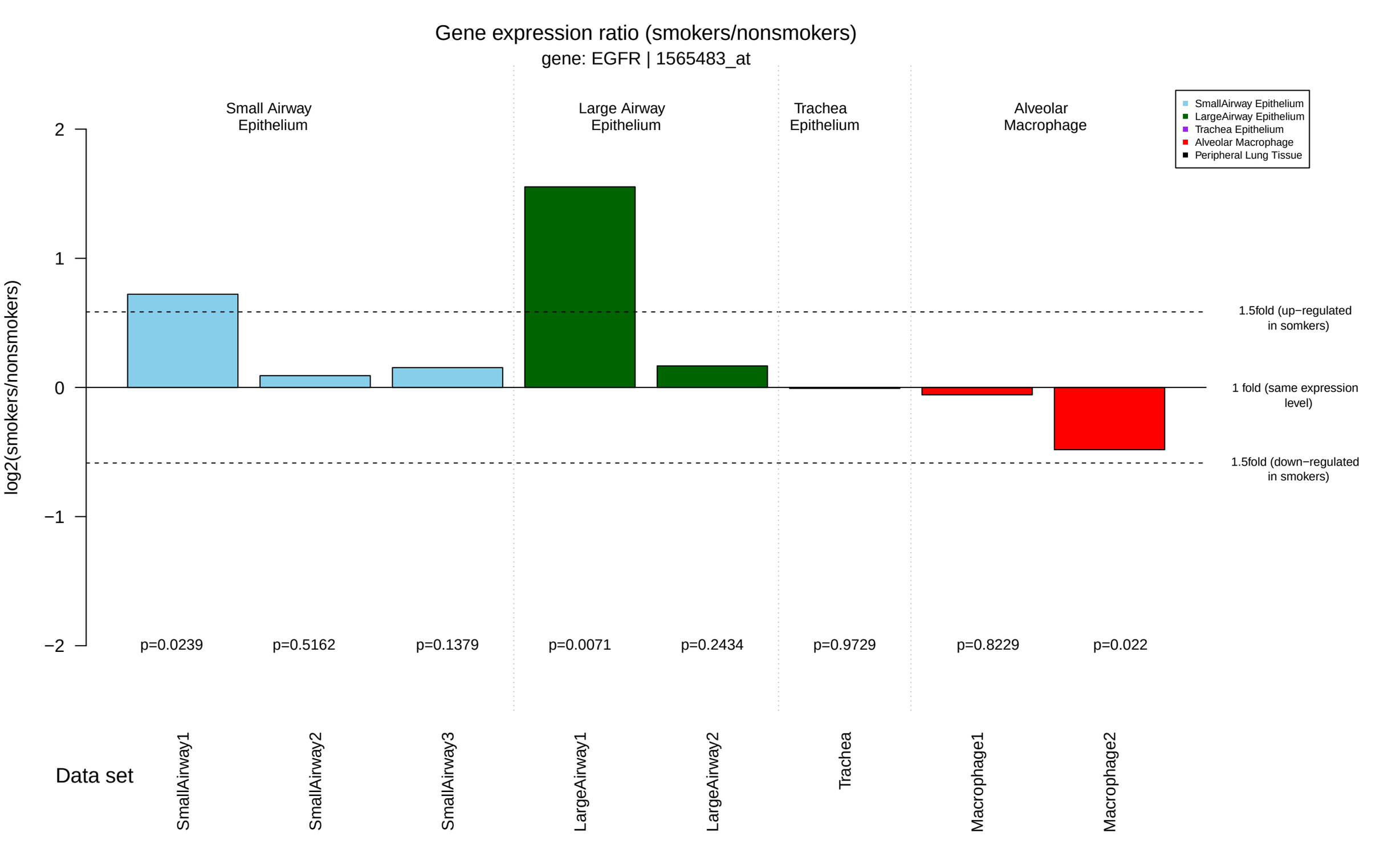
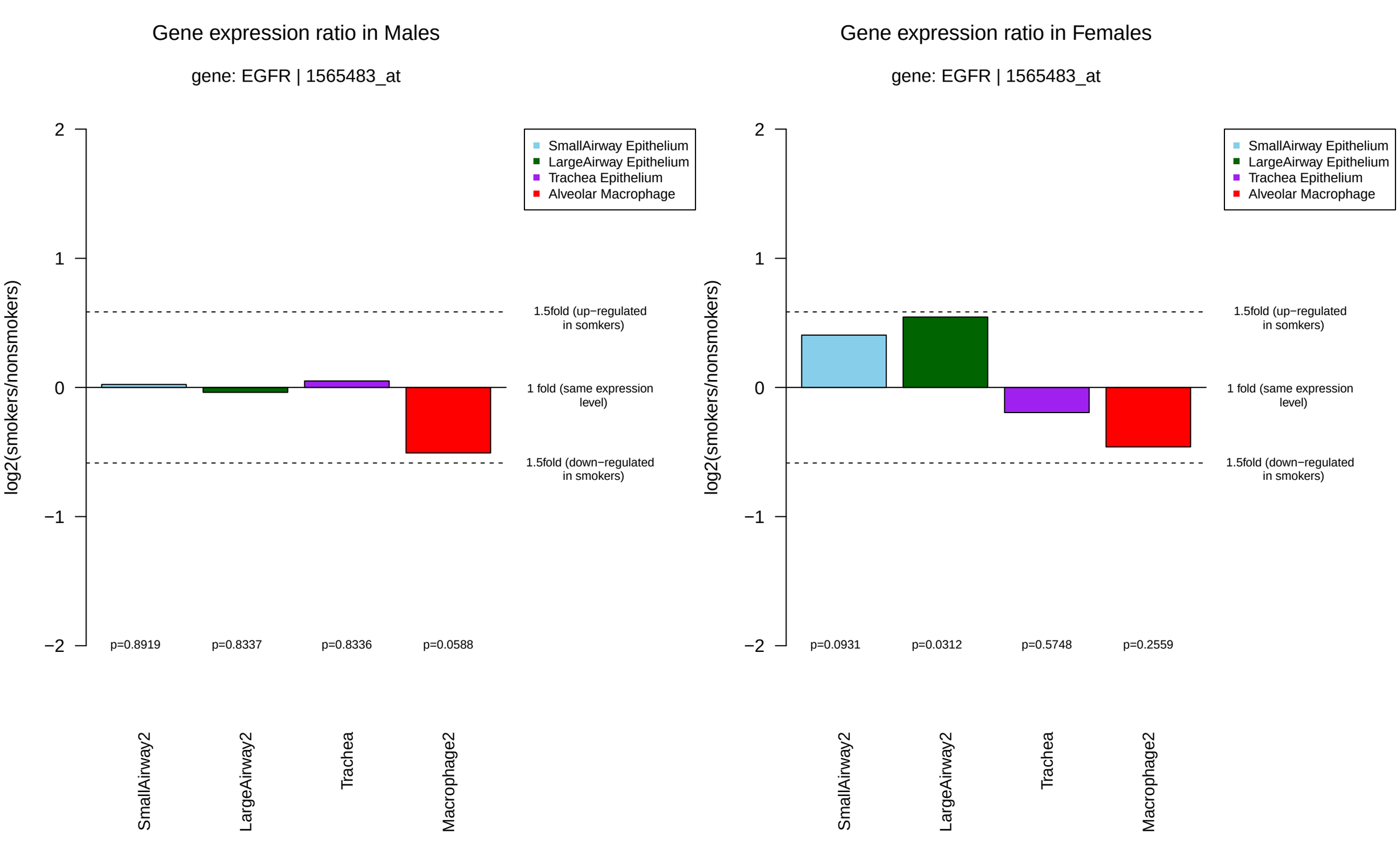
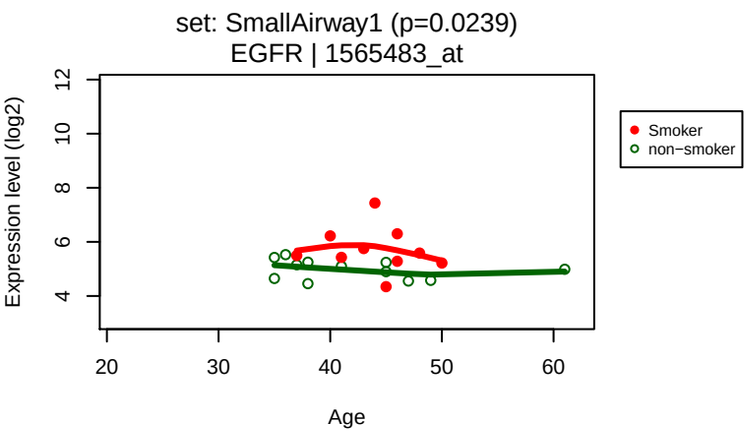
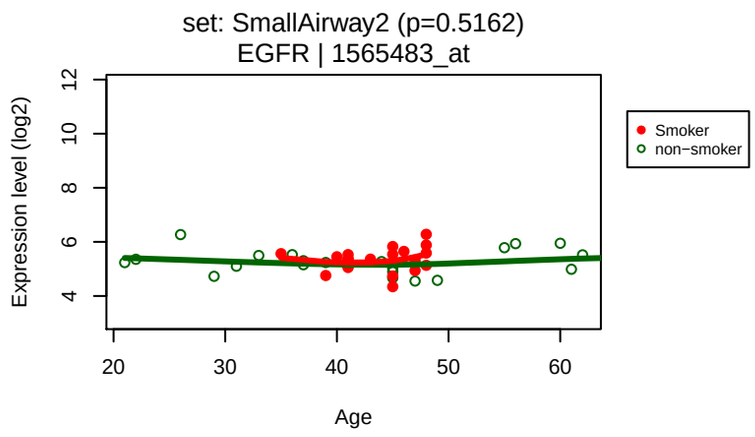
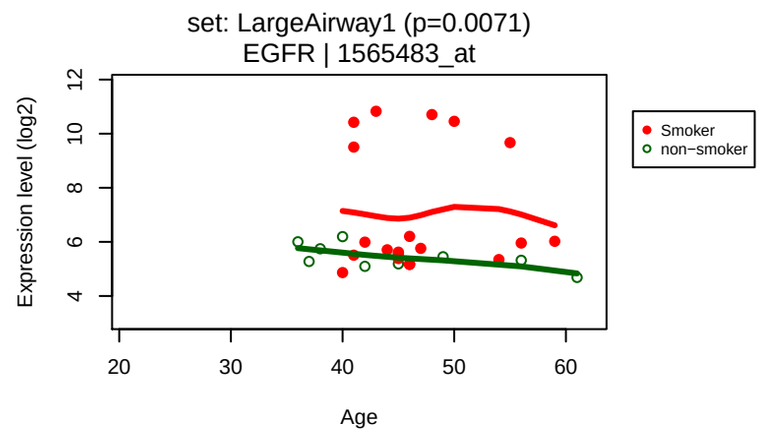

Gene expression of smokers and nonsmokers
gene: EGFR | 1565483_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0239
p=0.5162
p=0.1379
p=0.0071
p=0.2434
p=0.9729
p=0.8229
p=0.022
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

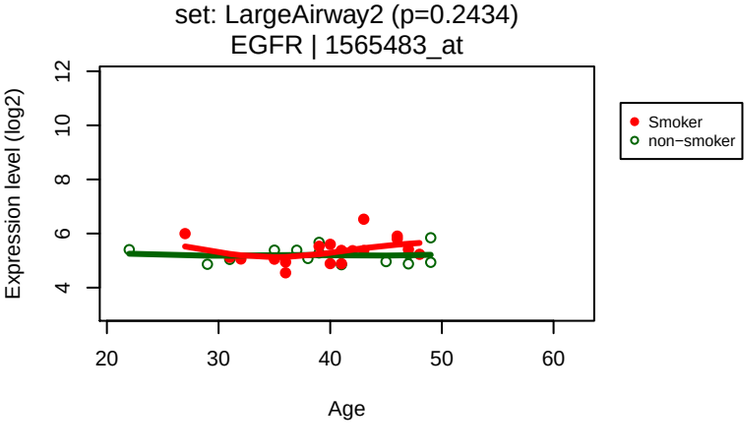

Gene expression ratio (smokers/nonsmokers)
gene: EGFR | 1565483_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0239
p=0.5162
p=0.1379
p=0.0071
p=0.2434
p=0.9729
p=0.8229
p=0.022
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

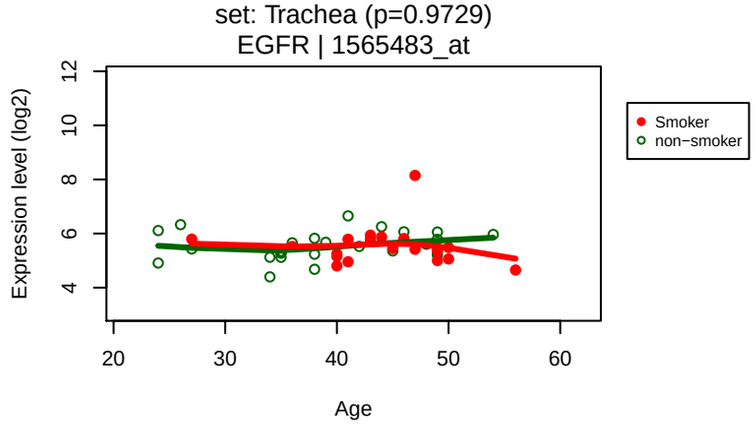

Gene expression ratio in Males
gene: EGFR | 1565483_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8919
p=0.8337
p=0.8336
p=0.0588
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: EGFR | 1565483_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0931
p=0.0312
p=0.5748
p=0.2559
SmallAirway2
LargeAirway2
Trachea
Macrophage2

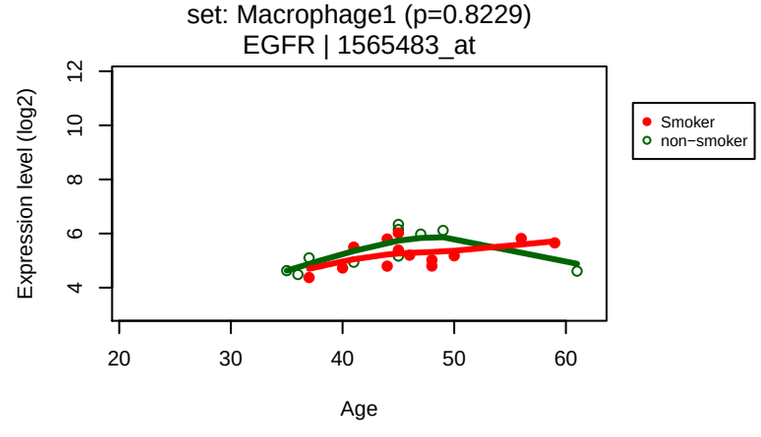

set: SmallAirway1 (p=0.0239)
EGFR | 1565483_at
Expression level (log2)
Age
Smoker
non−smoker

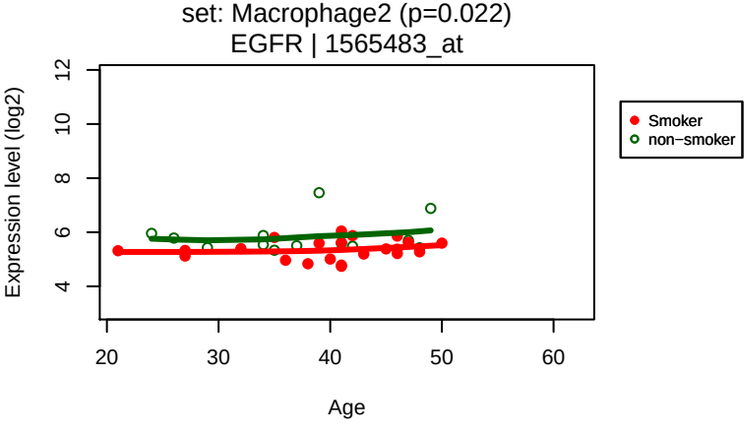

set: SmallAirway2 (p=0.5162)
EGFR | 1565483_at
Expression level (log2)
Age
Smoker
non−smoker

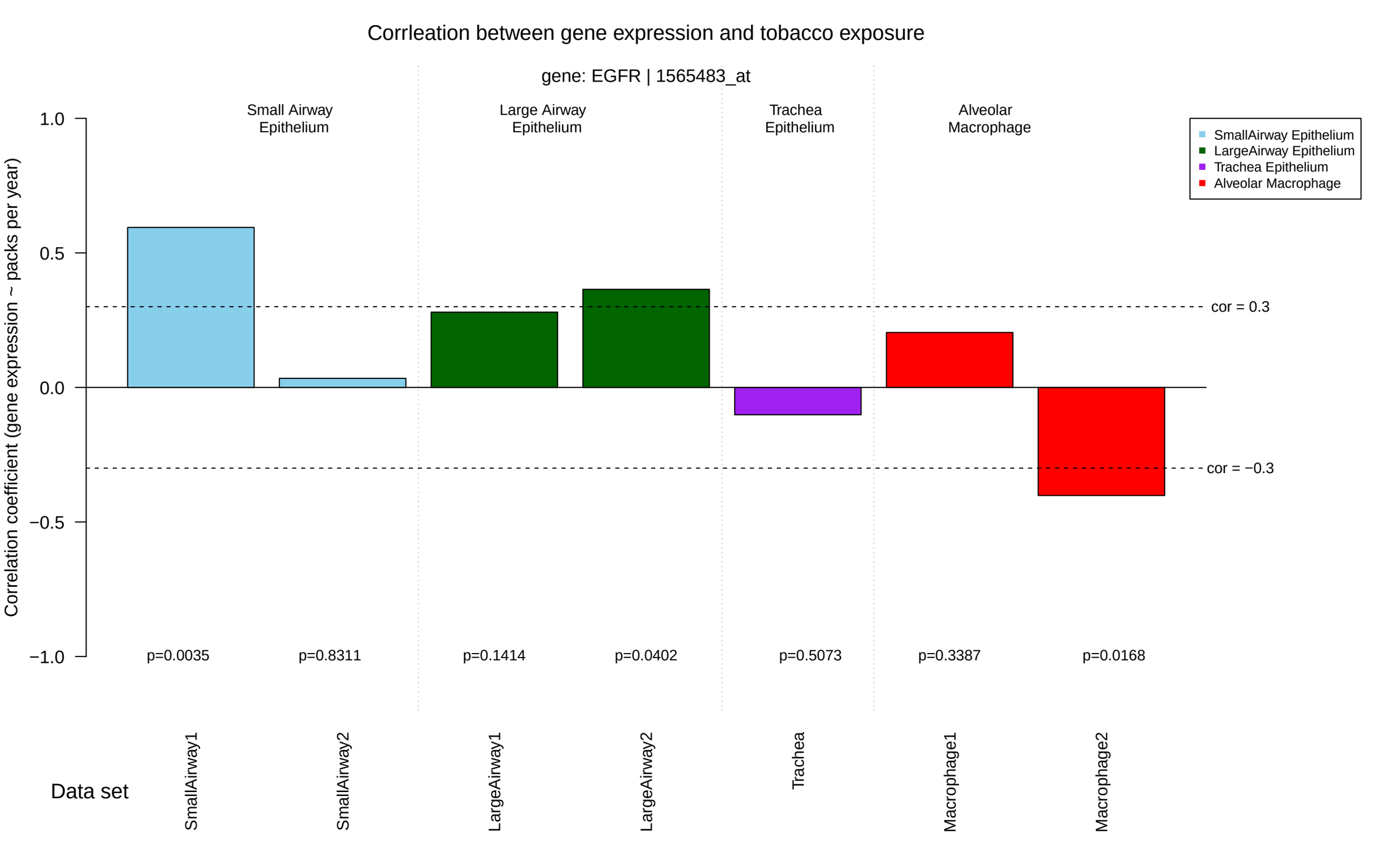

set: LargeAirway1 (p=0.0071)
EGFR | 1565483_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2434)
EGFR | 1565483_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9729)
EGFR | 1565483_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8229)
EGFR | 1565483_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.022)
EGFR | 1565483_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: EGFR | 1565483_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0035
p=0.8311
p=0.1414
p=0.0402
p=0.5073
p=0.3387
p=0.0168
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2